

ALEXANDRE CLIN DEFFARGES

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Education

Master's Thesis in LLM applied to education, MIT	Cambridge, US - Oct 2025 - Now
Master's degree in Robotic / AI, ETHZ	Zurich, Switzerland - 2023-2025
Global year exchange, Tokyo Institute of Technology	Tokyo, Japan - 2022 - 2023
Bachelor's degree in microtechnology, EPFL	Lausanne, Switzerland - 2020 - 2022

Work Experience

AI Specialist - Automating Student Essay Evaluation through LLMs - IBE | LLMs, Python

Zurich, Switzerland, 04/2024 - 07/2025, Remote

Alongside my other work, in my 40% part-time position at IBE, I dedicated my efforts to leveraging AI and LLMs to automate the grading of students' language essays and secondly to generate personalized exercises automatically.

Internship at SPOTS, Anti-Poaching Technology Development | Computer Vision, ROS2

Dinokeng game reserve, South Africa, 09/2024 - 12/2024, On site

At Dinokeng Game Reserve, I developed models for poacher detection via drone and an autonomous patrol car. I created models to detect humans and animals in video, raising alerts for unwanted presence, and worked on recognizing poacher footpaths.

Internship at Infosys, NLP and RAG Modeling | Deep Learning, RAG

Bengalore, India, 06/2024 - 08/2024, On site

I developed a model to classify and transform multimodal data (emails, PDFs, images, graphs) into structured data for a Retrieval-Augmented Generation (RAG) model, enhancing data processing and retrieval.

Internship at the Space Domain - Swiss Army | Signal processing, Machine learning, Github

Lausanne, Switzerland, 05/2023 - 08/2023, On site

During my internship at Solenix, I consulted on Swiss Army Space Domain projects, calculating satellite deltaV and fuel consumption using historical data, and applying signal processing and machine learning with Python and Java (<https://space-campus.ch/>).

Internship of manufacturing - ETML | Manufacturing

École technique École des métiers, Lausanne, 07/2022 - 08/2022, On site

During this internship, I learned drilling, milling, turning, brazing, and sawing by making various parts with different machines.

Projects and Research

LLM-Assisted Learning Protocols - MIT Media Lab | LLMs, HCI, Generative AI

Cambridge, US, 10/2025 - Now, On site

Developed a website with multiple LLM-powered educational modalities and a comprehensive research protocol to evaluate human-AI interaction. Identified strategies for integrating LLMs into the learning process to maximize student comprehension.

3D Tennis Player Reconstruction - MIT 100K Finalist | Computer Vision, 3D

Cambridge, US, 10/2025 - Now, On site

Selected as a finalist in the prestigious MIT 100K competition for developing a 3D reconstruction system for tennis players using 1-3 iPhones. This allows athletes to visualize matches from multiple angles, analyze mechanics, and enhance performance.

AI for Education: Transforming Learning Experiences - TalentKick Incubator

Zurich, Switzerland, 01/01/2024 - 10/2025

As part of the TalentKick incubator (<https://www.talentkick.ch/>), I built a team to develop an AI-powered education project that won the CHF 5,000 incubator prize. Our solution leverages AI to help students prepare for exams and enhances educational experiences.

Robotic Hand project - ETHZ | Robotic, Computer vision, CAD, RL

Zurich, Switzerland, 09/2023 - 01/2024, On site

Led 5-person team to build a CHF 250 robotic hand, integrating computer vision, RL and control for object manipulation.

Learning Recovery Maneuvers for Robotic Parkour - ETHZ | Reinforcement Learning

Zurich, Switzerland, 15/02/2024 - 15/06/2024, On site

Trained a Reinforcement Learning policy for the robot ANYmal to know how to fall and reduce its damage.

Spatial reasoning from LLM for global navigation - ETHZ | LLMs, python

Zurich, Switzerland, 15/02/2024 - 15/06/2024, On site

For my semester project, I tested the spatial reasoning capabilities of LLMs, such as GPT-4, to enhance robot navigation in very diverse environments. The aim was to evaluate the strengths and weaknesses of LLMs, followed by testing with the ANYmal robot.